

Lecture23_
MA5104

MA 5104: Statistical Inference and Learning

Module 3: Learning

Lecture 23: Loss, Risk, and What We Are Actually Estimating

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A Prediction Problem, 1801

On 1 January 1801 Giuseppe Piazzi observed a new object — Ceres — tracked it for about six weeks, and then lost it in the glare of the Sun.

Nobody knew where it would reappear.

A 24-year-old Carl Friedrich Gauss, who had made no observations of his own, fitted an orbit to the handful of noisy positions and said where to look. Ceres was recovered at the end of 1801, close to his prediction.

He could not observe the orbit. He could only predict the next observation. That is the whole subject.

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Prediction versus Inference

You have spent half a semester on inference, where the target was a parameter θ of a fixed population.

Inference

- Which predictors matter?
- How large is the effect, how sure are we?
- The form of f is part of the answer.
- A wrong model is a wrong conclusion.

Prediction

- What is Y at a new x ?
- Judged only on unseen data.
- f may be a black box.
- A wrong model can still predict well.

The same dataset can recommend different models depending on which of these you want.

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The Same Mathematics, A Different Target

Inference (what you have)	Learning (what we add)
parameter θ	function $f(\cdot)$
estimator $\hat{\theta}(D)$	fitted predictor $\hat{f}(\cdot; D)$
sampling distribution of $\hat{\theta}$	distribution of $\hat{f}(x)$ over training sets
$\text{MSE}(\hat{\theta}) = \text{bias}^2 + \text{var}$	pointwise $\text{MSE}(x)$: same identity
MLE: maximize $\sum_i \log p_{\theta}(y_i)$	ERM: minimize $\sum_i L(y_i, f(x_i))$
standard error	test error, cross-validated

Nothing ahead is a new *kind* of mathematics. It is the mathematics you already have, applied to a **function** instead of to a number.

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Loss and the Optimal Prediction

Prediction as a Decision Problem

At a given input x , the response Y is random with conditional distribution $P(Y | X = x)$. We must commit to a single number a .

A **loss function** $L(Y, a)$ says how costly it is to answer a when the truth turns out to be Y . The optimal answer minimizes conditional risk:

$$\hat{y}(x) = \arg \min_a \mathbb{E}[L(Y, a) | X = x]$$

The learning problem is defined by two things: the data-generating mechanism, and the loss. Change the loss and the correct answer changes, with the data untouched.

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Loss and the Optimal Prediction

Squared Loss Gives the Conditional Mean

Take $L(Y, a) = (Y - a)^2$ and let $f(x) = \mathbb{E}[Y | X = x]$. Add and subtract:

$$\mathbb{E}[(Y - a)^2 | X = x] = \mathbb{E}[(Y - f(x)) + (f(x) - a)]^2 | X = x]$$

The cross term is $2(f(x) - a) \mathbb{E}[Y - f(x) | X = x] = 0$, so

$$\mathbb{E}[(Y - a)^2 | X = x] = \underbrace{\text{Var}(Y | X = x)}_{\text{free of } a} + \underbrace{(f(x) - a)^2}_{\geq 0, = 0 \text{ iff } a = f(x)}$$

$f(x) = \mathbb{E}[Y | X = x]$ is optimal under squared loss.

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$$\begin{aligned} \mathbb{E}[(Y - f(x))^2 | X = x] &= \mathbb{E}[(Y - \mathbb{E}(Y|X=x) + \mathbb{E}(Y|X=x) - f(x))^2] \\ &= \mathbb{E}[(Y - \mathbb{E}(Y|X=x))^2] + \mathbb{E}[(\mathbb{E}(Y|X=x) - f(x))^2] \\ &\geq \text{Var}(Y | X = x) \\ &= \boxed{f(x) = \mathbb{E}[Y | X = x]} \end{aligned}$$

Loss and the Optimal Prediction

Absolute Loss Gives the Conditional Median

Now take $L(Y, a) = |Y - a|$. Differentiating the risk in a :

$$\frac{\partial}{\partial a} \mathbb{E}[|Y - a| | X = x] = P(Y < a | X = x) - P(Y > a | X = x)$$

Setting this to zero balances the two tails:

$$\hat{y}(x) = \text{median}(Y | X = x)$$

→ Errors grow linearly, so a single extreme Y no longer dominates.

→ **Historical note:** squares beat absolute deviations largely because they gave a closed form in an era without computers. That constraint is gone — choose your loss on purpose.

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Zero-One Loss Gives the Bayes Classifier

Let Y take values in $\{1, \dots, K\}$ and

$$L(Y, a) = \begin{cases} 0 & Y = a \\ 1 & Y \neq a \end{cases}$$

The conditional risk of answering a is $1 - P(Y = a | X = x)$, so

$$\hat{f}(x) = \arg \max_k P(Y = k | X = x)$$

This is the **Bayes classifier**; its error rate is the **Bayes rate**.

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Screening test.
X goes to a hospital if he is told that
prob. of him having cancer is 0.3.
What to do?

FN X has cancer, but you say NO cancer
CFN (Missed cancer) $\rightarrow C_{FN} \cdot p$

FP X has no cancer, but you say CANCER
CFP (cost of a scan) $\rightarrow C_{FP} \cdot (1-p)$

$$p = P(Y=1 | X=x)$$

say cancer $(1-p)C_{FP} < pC_{FN} ?$

Asymmetric Loss: Where the Threshold Really Comes From

Two classes, and the two mistakes are not equally bad: a false negative costs c_{FN} , a false positive costs c_{FP} . Write $p(x) = P(Y = 1 | X = x)$. Predicting 1 is better when

$$(1 - p(x)) c_{FP} < p(x) c_{FN}$$

$$\iff p(x) > \frac{c_{FP}}{c_{FP} + c_{FN}}$$

→ Equal costs give the familiar 0.5. Nothing else does.

→ A missed cancer at 100× the cost of a needless scan gives a threshold of $1/101 \approx 0.01$.

→ Same model, same data, same $p(x)$ — a completely different decision.

0.5 is not a law of probability. It is the answer to the question where both mistakes cost the same.

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Risk and the Irreducible Floor

For a predictor f , the **risk** is the expected loss over new data:

$$R(f) = \mathbb{E}_{(X,Y)}[L(Y, f(X))]$$

Write f^* for the minimizer over *all* functions — the conditional mean, median, or Bayes classifier, depending on L .

With $Y = f^*(X) + \epsilon$ and squared loss, for any fitted $\hat{f}$:

$$\mathbb{E}[(Y - \hat{f}(x))^2 | X = x] = \underbrace{(f^*(x) - \hat{f}(x))^2}_{\text{reducible}} + \underbrace{\sigma^2}_{\text{irreducible}}$$

No method, however flexible, and no amount of data, touches σ^2 .

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Y, X
 $Y = f^*(X) + \epsilon$
 $\hat{f}(x) \rightarrow$ given data, model
 $F \leftarrow$ family of functions
 $\hookrightarrow$ Linear Models
 $\hookrightarrow$ Random forests (Decision Trees)
 $\hookrightarrow$ Neural Networks (NN)
 $\hat{f} \rightarrow F \rightarrow f^*$
 $f_1, f_2, \dots$

Excess Risk: Approximation plus Estimation

We never optimize over all functions. We pick a class $\mathcal{F}$ — linear, sparse, additive, trees, networks — and find $\hat{f} \in \mathcal{F}$ from data. Add and subtract $\inf_{f \in \mathcal{F}} R(f)$:

$$R(\hat{f}) - R(f^*) = \underbrace{\left(\inf_{f \in \mathcal{F}} R(f) - R(f^*) \right)}_{\text{approximation error}} + \underbrace{\left(R(\hat{f}) - \inf_{f \in \mathcal{F}} R(f) \right)}_{\text{estimation error}}$$

Both brackets are non-negative.

→ **Approximation**: how wrong is the *best* member of $\mathcal{F}$? Shrinks as $\mathcal{F}$ grows. Does not depend on n .

→ **Estimation**: how far is the member we *found* from the best one? Grows as $\mathcal{F}$ grows; shrinks as n grows.

Enlarging $\mathcal{F}$ buys the first term and pays for it with the second. This is the bias-variance trade, stated for *function classes*.

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Every Method Is a Choice of $\mathcal{F}$

Restriction on $\mathcal{F}$	Method	Lecture
linear in x	least squares, logistic	3, 4
linear and <i>sparse</i>	subset selection, lasso	7, 8
linear, shrunk	ridge, elastic net	8
additive, smooth in each x_j	splines, GAMs	9
piecewise constant on boxes	trees	10
averages of many such	forests, boosting	11, 12
large margin in a feature space	SVM	13
compositions of simple maps	neural networks	14

For each row, the two questions are always the same: how much approximation error does the restriction cost, and how much estimation error does it save?

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Case Study 1: Spam versus Regular Email

TABLE 1.1. Average percentage of words or characters in an email message equal to the indicated word or character. We have chosen the words and characters showing the largest difference between *spam* and *email*.

	george	you	your	hp	free	hpl	!	our	re	edu	remove
spam	0.00	2.26	1.38	0.02	0.52	0.01	0.51	0.51	0.13	0.01	0.28
email	1.27	1.27	0.44	0.90	0.07	0.43	0.11	0.18	0.42	0.29	0.01

Looking only at this table — no tools yet — what decision rules would you write down?

And: why is *george* the most discriminating word in a spam dataset?

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Two Hand-Built Learners, and a Warning

$$\begin{aligned}
 & \text{hit} \rightarrow (\% \text{george} < 0.6) \text{ and } (\% \text{you} > 1.5) \Rightarrow \text{spam} \\
 & \quad \quad \quad 0.2\% \text{you} - 0.3\% \text{george} > 0 \Rightarrow \text{spam}
 \end{aligned}$$

$\sum \beta_i x_i$

You have just invented two function classes: the first partitions space into *boxes* (a tree, L10), the second thresholds a *weighted sum* (a linear classifier, L3–L4).

And the warning

These data came from one employee's mailbox at HP Labs. *george* and *hp* are features of *that inbox*, not of spam. The model learned the data collection, not the phenomenon — and the training error would never have told you.

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Case Study 2: Prostate Cancer

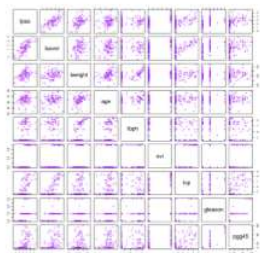


FIGURE 1.1. Scatterplot matrix of the prostate cancer data. The first row shows the response against each of the predictors in turn. Two of the predictors, *lvi* and *lgleason*, are categorical.

Stamey et al. (1989): PSA and clinical measures on 97 men before radical prostatectomy.

Goal: predict *lpsa* from *lcavol*, *lweight*, *age*, *lbph*, and others. Continuous response $\Rightarrow$ **regression**.

Some correlations are visible. A good predictive model is *not* constructible by eye — the plot tells you what kind of problem you have, not how to solve it.

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Exploratory Data Analysis

EDA is basic statistics applied to every column, plus pictures: means, medians, standard deviations, histograms, boxplots, scatter plots.

What you are hunting for: **hidden outliers**, **skew**, **class imbalance**, **misleading correlations**, and data-quality failures — before they become modelling failures.

A workable order: understand the actual problem $\rightarrow$ inspect $\rightarrow$ visualize $\rightarrow$ then choose a loss and a class.

Plots are tools for thinking, not decoration for reporting. Make them complete, and make them readable by color-blind viewers.

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Four Datasets

Four datasets of eleven (x, y) pairs.

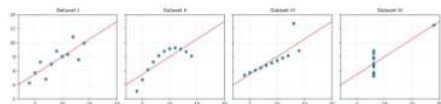
Summary	Value, all four
Mean of x	9.00
Sample variance of x	11.00
Mean of y	7.50
Sample variance of y	4.13
Correlation	0.816
Least-squares line	$y = 3.00 + 0.500x$

Every number a summary would print is identical to two decimals.

Before I show you the plots: how differently could these possibly look?

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Anscombe's Quartet (1973)



A clean linear relation; a curve; a linear relation destroyed by one outlier; a vertical stack plus one leverage point manufacturing the entire correlation.

Same lesson as the noise demonstration: there, one number looked excellent over nothing; here, six numbers look identical over four different truths.

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Case Study 3: Handwritten Digits



FIGURE 1.2. Examples of handwritten digits from U.S. postal envelopes.

Classification, with a very low tolerable error rate; ambiguous items get routed to a "don't know" pile and sorted by hand.

Trained human readers still misclassify a few percent of these characters.

Also note: each observation lives in $\mathbb{R}^{256}$. There is no scatterplot matrix here. What happens to our methods when p is large is Lecture 2.

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Things to Take Away

Bayes loss squared - error loss

- Which function is optimal depends on the loss you choose — mean, median, Bayes classifier, or a shifted threshold. The data alone do not determine the answer.
- No method touches the Bayes risk. Beating it is evidence of a leak, not of a discovery.
- Excess risk = approximation + estimation. Choosing a method is choosing a function class, and enlarging the class trades one term against the other.

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